How the AI Works: Step-by-Step Execution Pipelines

Exact flow of Q&A (RAG), Document Summaries, Clause Analysis & Blind-Spot Detection

1. How Q&A Works (RAG Pipeline)

Ingestion & Vector Indexing Phase:

Your Legal PDF / Document (or scanned image auto-converted via OCR)

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Split into **~800-character semantic chunks** (preserving page numbers)

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Each chunk → converted to a **384-dimensional number vector (embedding)** via HuggingFace *all-MiniLM-L6-v2*

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All vectors saved to **FAISS** (local vector database stored on your disk)

[PHASE 2: WHEN YOU ASK A QUESTION]

Your question → also converted to a 384-dimensional vector

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FAISS finds the top 5 most similar chunks by mathematical vector distance (cosine similarity)

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Those 5 chunks + your question → sent to the LLM (Groq / Gemini) as a strict prompt: *"Answer ONLY from these provided excerpts: [chunks]. Question: [your question]"*

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AI generates the answer → returned immediately with exact clickable **page sources**

Key Point: The LLM **never reads the entire PDF blindly**. It only receives the 5 most relevant chunks retrieved by FAISS. This prevents hallucinations and guarantees pinpoint page citations.

2. How Summarisation Works

Your PDF

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All pages extracted & joined into one continuous text (capped at 15,000 characters for optimal context)

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Sent to the LLM with a strict JSON-enforcement prompt:
"Read this legal agreement. Return a structured JSON containing: parties, effective dates, core obligations, financial terms, risk flags, governing law, and termination conditions."

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LLM returns structured JSON data (validated against Pydantic schema)

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Frontend dynamically renders each field into dedicated UI cards: Executive Overview, Commercial Terms, and Practical Takeaways

Key Point: Zero unstructured guesswork. Strict JSON formatting ensures consistent commercial extraction every single time, without missing dates, parties, or dollar amounts.

3. How Clause Analysis & Risk Profiling Works

Your PDF

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System scans text to locate standard contract clauses (Indemnification, Termination, Governing Law, Liability Cap, Confidentiality, Dispute Resolution)

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Relevant clause excerpts sent to the LLM with an audit rubric:
"Classify each clause, evaluate balance of power, assign Risk (High / Medium / Low), and give risk rationale."

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AI evaluates legal risk:
• **HIGH RISK:** Unlimited liability, one-sided indemnification, automatic renewals with short notice.
• **MEDIUM RISK:** Ambiguous termination triggers or unfamiliar foreign governing jurisdictions.
• **LOW RISK:** Standard bilateral confidentiality and mutual dispute resolution.

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Returns structured list of clauses with severity color badges, exact quotes, and proposed amendment language

Key Point: Surfaces dangerous legal commitments *before* you sign, giving you the exact rationale and redline suggestions to negotiate with the counterparty.

4. How Contract Blind-Spot Detector Works

Your PDF

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Full document text extracted and audited against a comprehensive **Contract Gap Checklist** (liability limits, remedies for breach, unattached exhibits, undefined terms, vague deadlines)

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Sent to the LLM with a specialized omission-detection prompt:
"Identify what is MISSING, VAGUE, or UNRESOLVED. Check for: 1) Potential Gaps, 2) Ambiguous terms, 3) Missing references, 4) Unresolved contingencies. Do not hallucinate."

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AI classifies findings into 4 distinct gap categories:
• **Potential Gap:** Missing protective provisions (e.g. no cap on damages, no audit rights).
• **Ambiguity:** Subjective phrases like *"reasonable efforts"*, *"timely manner"*, or *"to satisfaction"*.
• **Missing Reference:** Refers to unattached schedules (e.g. *"as defined in Exhibit B"*, but Exhibit B is missing).
• **Needs Human Review:** Unilateral obligations or assumptions requiring lawyer clarification.

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Outputs comprehensive audit card for each finding:
Severity Badge (High / Med / Low) + Explanation + Supporting Quote + Page Number + **Targeted Question for Human Counsel**

Key Point: Unlike standard tools that only read what is written on the page, the **Blind-Spot Detector actively searches for what was left out**. It equips legal counsel with specific questions to ask the counterparty during negotiation.

Architecture Summary:
• **Q&A:** Local FAISS vector search + 5 retrieved chunks + strict LLM prompt + page citations.
• **Summary:** Full-text ingestion + structured JSON schema prompt + card-based rendering.
• **Clause Analysis:** Pattern extraction + risk rubric (High/Med/Low) + amendment recommendations.
• **Blind-Spot Detector:** Omission auditing + 4-gap taxonomy + human counsel questions.